RAJALAKSHMI ENGINEERING COLLEGE

Rakesh Kumar S 251001205-2025-IT

R2

Started on	Friday, 7 August 2026, 9:32 AM
State	Finished
Completed on	Friday, 14 August 2026, 9:00 AM
Time taken	6 days 23 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n, i;
    int counter = 0;

    scanf("%d", &n);

    for (i = 1; i <= n; ++i)
    {
        counter++;              // for-loop condition

        if (n % i == 0)
        {
            counter++;          // printf("%d ", i)
        }

        counter++;              // if condition
    }

    counter++;                  // final failed for-loop condition

    printf("%d", counter);

    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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